SCPT Engineering Interpretation Report

Forward ray-path shear-wave velocity inversion, arrival-time comparison, and Vs30 summary

Report generated	2026-08-05 11:46 New Zealand Standard Time
Application version	0.3.0-alpha.1
Project schema	5
Project	Untitled
GRU source	18563_6.GRU
Source offset	2.400 m
Selected detailed model	Pair crossover
Smoothing / regularisation	0.35
Vs30 extrapolation weighting	1.00
Deepest receiver	37.47 m
GRU pre-trigger correction	50.0 ms
Waveform review	74 accepted; 0 rejected/excluded; 0 not reviewed
QC warning thresholds	SNR < 10 dB; sign-reversed correlation < 0.60; PT disagreement > max(2 ms, two samples); sample-interval variation > 1%

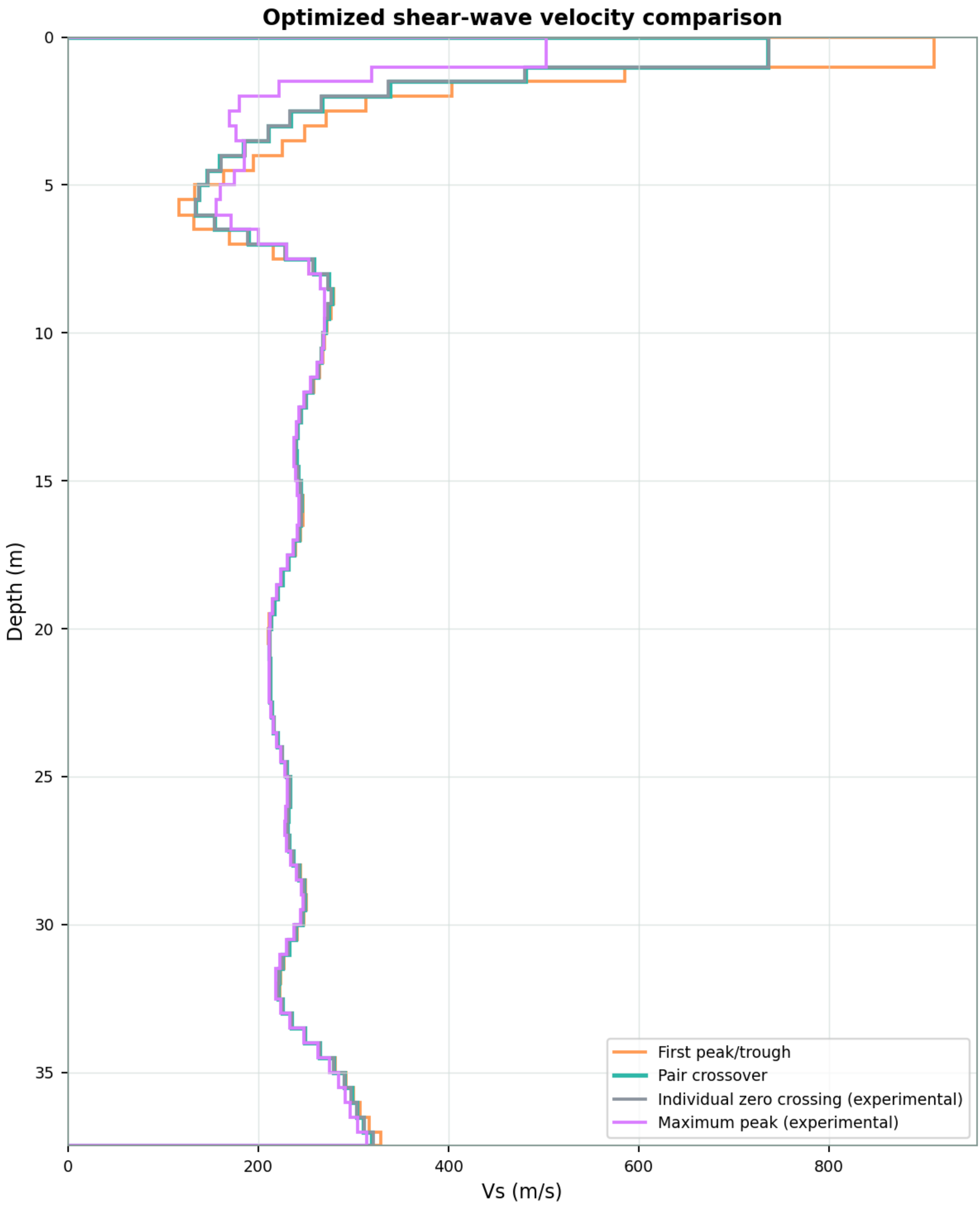
Model comparison

Pick definition	Layers	RMSE (ms)	Vs30 (m/s)	Extrapolation
First peak/trough	74	0.679	234.0	None - measured to 30 m
Pair crossover	74	0.626	231.3	None - measured to 30 m
Individual zero crossing (experimental)	74	0.621	231.3	None - measured to 30 m
Maximum peak (experimental)	74	0.332	227.2	None - measured to 30 m

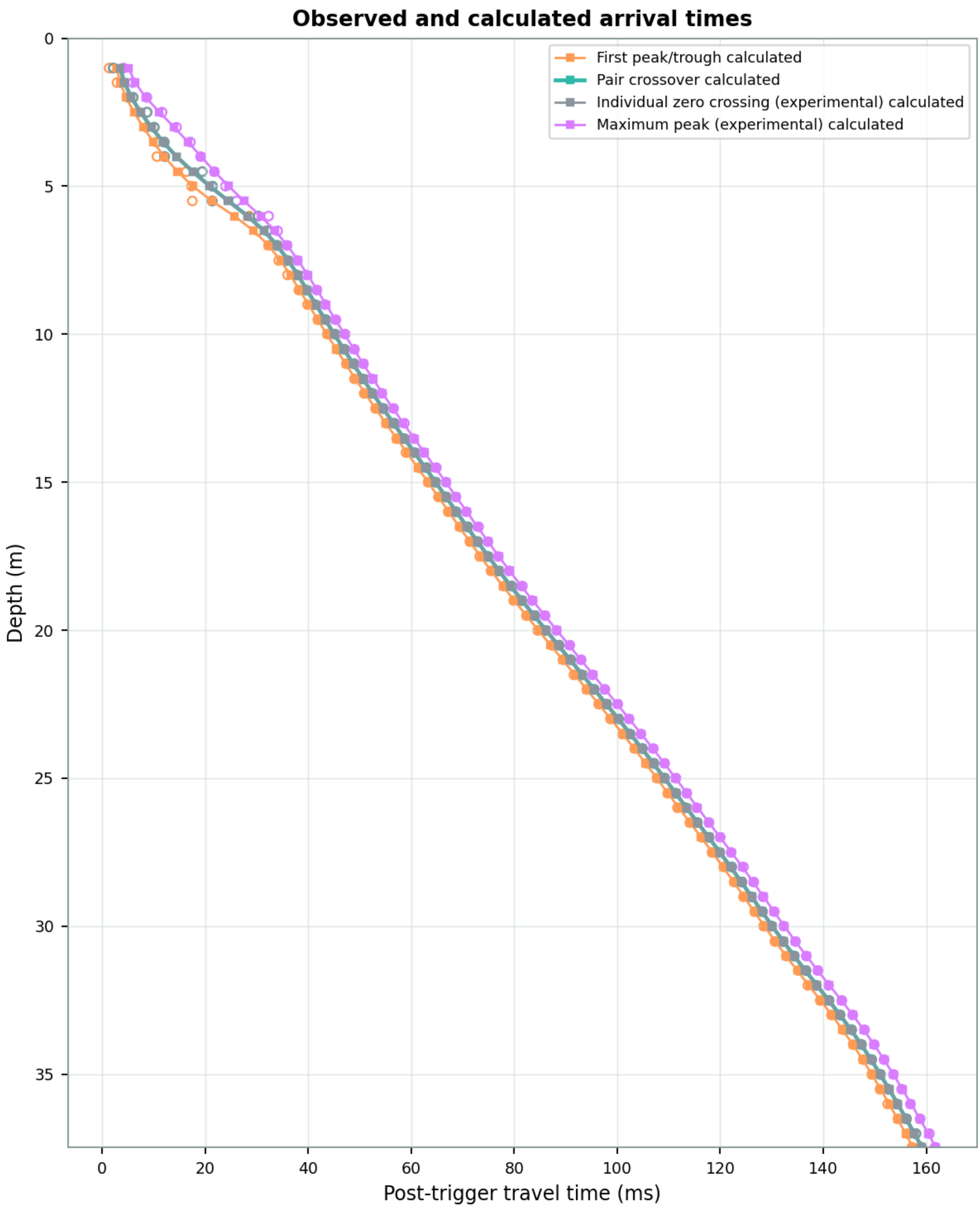
Arrival definitions: First peak/trough is the mean of the reviewed left and right extrema; Pair crossover is one reviewed time where the reversed traces intersect after arrival; Individual zero crossing is the mean of two per-trace zero-axis crossings and is experimental; Maximum peak is the mean of two maximum-amplitude times and is experimental.

Vs30 is calculated as 30 divided by the summed vertical shear-wave travel time. Profiles between 25 m and 30 m use the configured interval weighting only to estimate the missing depth. Automatic waveform suggestions should be reviewed by a qualified operator before relying on this report.

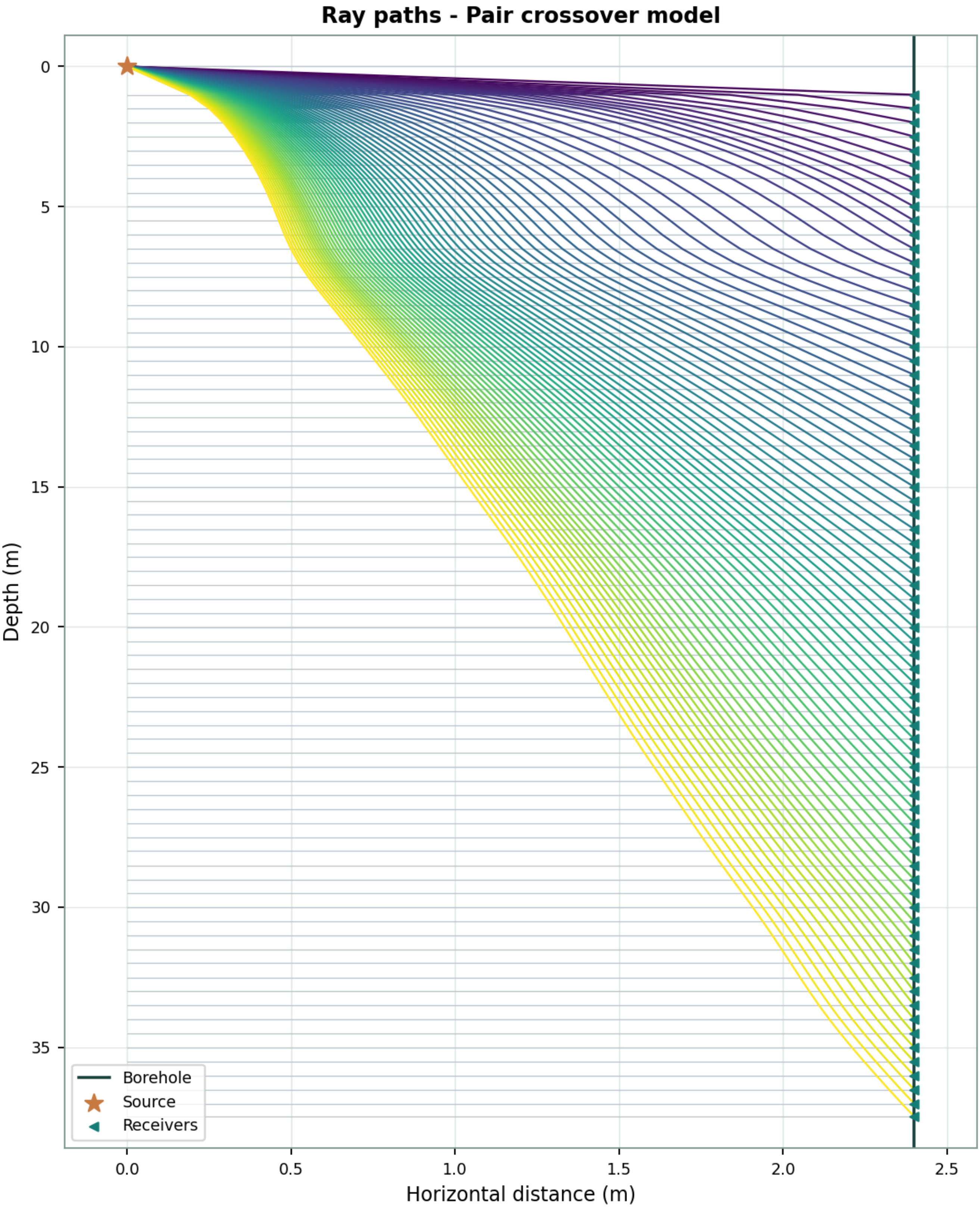
Velocity profile comparison



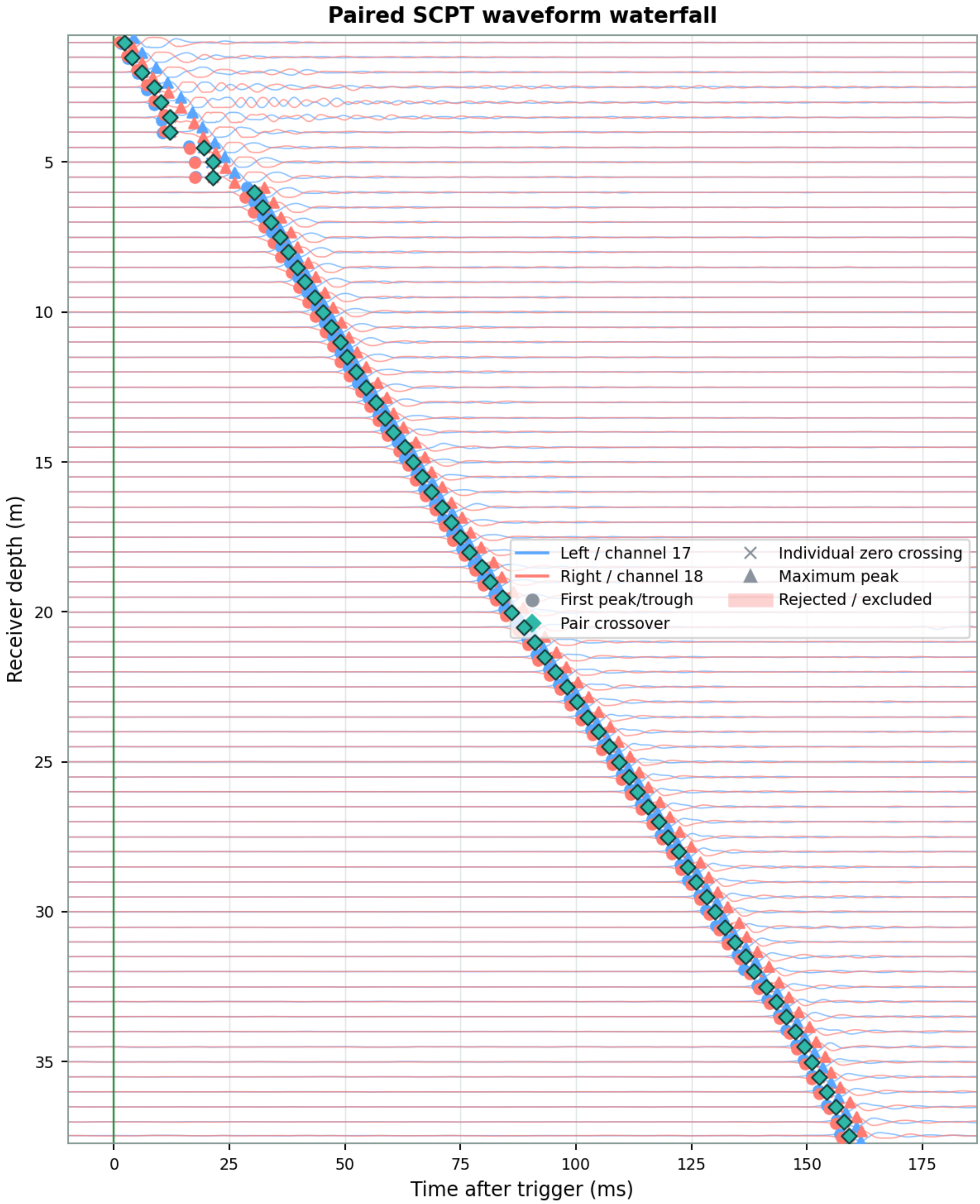
Arrival-time fit comparison



Selected-model ray paths



Waveform waterfall and reviewed picks



Receiver pick times

Depth (m)	First peak/trough (ms)	Pair crossover (ms)	Individual zero crossing (experimental) (ms)	Maximum peak (experimental)
1.010	1.380	2.200	2.153	4.161
1.500	2.921	3.808	3.808	5.986
2.000	4.998	6.048	6.064	8.711
2.500	7.112	8.752	8.776	11.664
3.010	8.646	10.155	10.143	14.469
3.510	10.531	12.037	12.100	17.155
4.000	10.656	12.145	12.161	19.133
4.510	16.350	19.487	19.408	21.817
5.000	17.382	21.470	21.375	23.974
5.500	17.535	21.385	21.385	26.123
6.010	28.580	30.339	30.307	32.347
6.510	30.274	32.080	32.064	34.093
7.000	32.327	33.975	33.975	35.908
7.510	34.300	35.892	35.892	37.900
8.000	36.032	37.741	37.740	39.677
8.510	38.202	39.675	39.675	41.700
9.000	39.897	41.244	41.291	43.300
9.510	41.900	43.436	43.439	45.400
10.000	43.722	45.241	45.241	47.200
10.510	45.724	47.039	47.118	49.000
11.000	47.444	48.934	48.965	50.756
11.510	49.000	50.471	50.474	52.400
12.000	50.900	52.410	52.420	54.300
12.510	53.100	54.569	54.577	56.600
13.010	55.200	56.718	56.725	58.700
13.530	57.200	58.663	58.664	60.477
14.000	59.000	60.449	60.456	62.400
14.510	61.500	63.000	63.020	64.900
15.000	63.300	64.741	64.750	66.800
15.500	65.300	66.751	66.752	68.700
16.000	67.200	68.694	68.693	70.700
16.510	69.493	70.992	70.993	73.100
17.010	71.400	72.908	72.909	74.900
17.510	73.300	74.885	74.887	76.900
18.000	75.500	76.946	76.945	79.000
18.510	78.000	79.557	79.571	81.600
19.000	79.900	81.422	81.427	83.500
19.510	82.400	83.979	83.996	86.000
20.010	84.569	86.041	86.052	88.200

Depth (m)	First peak/trough (ms)	Pair crossover (ms)	Individual zero crossing (experimental) (ms)	Maximum peak (experimental)
20.510	87.300	88.704	88.703	90.800
21.000	89.500	90.990	90.992	93.000
21.510	91.600	93.100	93.115	95.200
22.000	94.100	95.519	95.522	97.600
22.500	96.400	97.926	97.945	100.100
23.000	98.700	100.249	100.255	102.300
23.510	101.000	102.446	102.442	104.600
24.000	103.400	104.882	104.897	107.000
24.500	105.700	107.104	107.104	109.200
25.000	107.700	109.207	109.207	111.400
25.510	109.800	111.363	111.363	113.500
26.000	111.700	113.243	113.242	115.400
26.500	114.100	115.530	115.534	117.800
27.000	116.400	117.812	117.812	120.000
27.510	118.400	119.872	119.874	122.100
28.000	120.700	122.223	122.223	124.500
28.510	122.700	124.199	124.201	126.500
29.010	124.500	125.990	125.991	128.300
29.510	126.700	128.185	128.183	130.500
30.000	128.400	129.972	129.984	132.300
30.520	130.600	132.199	132.203	134.600
31.010	132.800	134.355	134.355	136.700
31.500	135.158	136.647	136.650	138.968
32.000	137.000	138.555	138.606	141.000
32.510	139.400	141.092	141.098	143.600
33.000	141.600	143.234	143.247	145.700
33.500	143.900	145.492	145.489	148.000
34.000	145.800	147.376	147.408	149.900
34.510	147.747	149.395	149.379	151.788
35.010	149.400	151.074	151.088	153.600
35.510	151.000	152.607	152.607	155.200
36.010	152.500	154.252	154.255	156.900
36.510	154.500	156.109	156.157	158.800
37.010	156.300	157.965	157.966	160.600
37.470	157.200	158.989	158.986	161.700

Waveform QC and exclusions schedule

QC metrics are deterministic review aids. A warning does not automatically reject an observation. Only records explicitly marked Rejected are excluded from inversion. Recorded arrival uncertainty is audit information at this development stage and does not yet weight the inversion.

Depth	Review state	Unc.	SNR L/R	Noise RMS L/R	Corr.	Lag	Delta PT/Z/M	Polarity	Warnings
1.01	Accepted	0.100	47.6/42.7	0.0363/0.0642	0.979	+0.20	0.3/0.2/0.2	Yes	None
1.50	Accepted	0.100	46.7/48.9	0.0412/0.0321	0.982	+0.20	0.2/0.3/0.1	Yes	None
2.00	Accepted	0.100	48.8/46.4	0.0324/0.0424	0.953	-0.20	0.3/0.2/0.7	Yes	None
2.50	Accepted	0.100	41.0/39.0	0.079/0.0998	0.966	+0.20	0.0/0.1/0.1	Yes	None
3.01	Accepted	0.100	42.9/43.3	0.0637/0.0612	0.967	+0.20	0.1/0.5/0.1	Yes	None
3.51	Accepted	0.100	42.6/45.1	0.0657/0.0496	0.975	+0.40	0.5/0.4/0.4	Yes	None
4.00	Accepted	0.100	40.0/39.2	0.0895/0.0974	0.969	+0.80	0.3/0.2/0.1	Yes	None
4.51	Accepted	0.100	43.7/43.9	0.0579/0.0568	0.979	+0.40	0.2/0.3/0.1	Yes	None
5.00	Accepted	0.100	39.3/44.3	0.0962/0.0542	0.985	+0.20	0.0/0.5/0.1	Yes	None
5.50	Accepted	0.100	42.2/46.1	0.0687/0.0443	0.981	+0.40	0.2/0.0/0.0	Yes	None
6.01	Accepted	0.100	42.0/43.0	0.0709/0.0626	0.986	+0.20	0.2/0.0/0.3	Yes	None
6.51	Accepted	0.100	42.8/42.8	0.0641/0.0645	0.979	+0.40	0.1/0.4/0.6	Yes	None
7.00	Accepted	0.100	39.5/38.0	0.0941/0.112	0.976	+0.40	0.2/0.4/0.4	Yes	None
7.51	Accepted	0.100	44.7/41.8	0.0516/0.0714	0.972	+0.40	0.2/0.3/0.6	Yes	None
8.00	Accepted	0.100	43.5/43.5	0.0596/0.0587	0.988	+0.00	0.1/0.0/0.1	Yes	None
8.51	Accepted	0.100	42.4/42.8	0.0671/0.0636	0.969	+0.60	0.5/0.4/0.6	Yes	None
9.00	Accepted	0.100	43.1/42.6	0.0622/0.0649	0.971	+0.40	0.2/0.3/0.6	Yes	None
9.51	Accepted	0.100	44.2/45.0	0.0547/0.0491	0.983	+0.40	0.2/0.3/0.4	Yes	None
10.00	Accepted	0.100	42.2/40.7	0.0685/0.0807	0.981	+0.20	0.2/0.1/0.4	Yes	None
10.51	Accepted	0.100	41.5/42.2	0.0747/0.068	0.993	+0.20	0.2/0.2/0.4	Yes	None
11.00	Accepted	0.100	43.8/40.8	0.0575/0.0795	0.973	+0.20	0.0/0.1/0.0	Yes	None
11.51	Accepted	0.100	41.8/40.3	0.0716/0.0851	0.971	+0.40	0.0/0.3/0.4	Yes	None
12.00	Accepted	0.100	42.1/42.3	0.0692/0.0673	0.985	+0.40	0.2/0.4/0.6	Yes	None
12.51	Accepted	0.100	45.7/40.6	0.0461/0.0822	0.988	+0.80	0.6/0.7/0.8	Yes	None
13.01	Accepted	0.100	41.1/42.5	0.0784/0.0662	0.991	+0.60	0.4/0.5/0.6	Yes	None

Depth	Review state	Unc.	SNR L/R	Noise RMS L/R	Corr.	Lag	Delta PT/Z/M	Polarity	Warnings
13.53	Accepted	0.100	42.1/42.1	0.0687/0.0678	0.990	+0.00	0.0/0.0/0.1	Yes	None
14.00	Accepted	0.100	43.3/41.9	0.0605/0.0701	0.984	+0.60	0.4/0.4/0.4	Yes	None
14.51	Accepted	0.100	35.5/45.7	0.0693/0.0454	0.846	+0.00	0.2/0.4/0.6	Yes	None
15.00	Accepted	0.100	44.4/40.5	0.0527/0.0831	0.982	+0.80	0.6/0.7/0.8	Yes	None
15.50	Accepted	0.100	41.1/44.7	0.0763/0.0503	0.988	+0.00	0.2/0.2/0.2	Yes	None
16.00	Accepted	0.100	45.0/41.7	0.0475/0.0727	0.976	+0.40	0.4/0.4/0.6	Yes	None
16.51	Accepted	0.100	36.5/43.4	0.0795/0.0585	0.973	-0.40	0.2/0.1/0.2	Yes	None
17.01	Accepted	0.100	40.5/42.5	0.0796/0.066	0.984	+0.40	0.4/0.4/0.6	Yes	None
17.51	Accepted	0.100	41.9/41.9	0.0666/0.0692	0.994	+0.20	0.2/0.2/0.2	Yes	None
18.00	Accepted	0.100	42.2/40.8	0.0662/0.0792	0.972	+0.60	0.6/0.6/0.8	Yes	None
18.51	Accepted	0.100	40.8/39.6	0.0776/0.0913	0.987	+0.60	0.4/0.5/0.8	Yes	None
19.00	Accepted	0.100	42.2/41.0	0.0645/0.0776	0.975	+0.60	0.2/0.5/0.6	Yes	None
19.51	Accepted	0.100	44.9/40.2	0.0486/0.0849	0.972	+0.80	0.4/0.6/0.8	Yes	None
20.01	Accepted	0.100	42.2/42.3	0.0639/0.0674	0.980	+0.80	0.5/0.6/0.8	Yes	None
20.51	Accepted	0.100	37.1/40.0	0.0785/0.0878	0.987	+0.40	0.2/0.4/0.4	Yes	None
21.00	Accepted	0.100	41.1/43.3	0.0754/0.0597	0.986	+0.40	0.2/0.3/0.4	Yes	None
21.51	Accepted	0.100	41.5/40.1	0.0721/0.0862	0.981	+0.80	0.4/0.7/0.8	Yes	None
22.00	Accepted	0.100	40.2/43.4	0.0832/0.0593	0.989	+0.40	0.2/0.3/0.4	Yes	None
22.50	Accepted	0.100	40.2/40.9	0.0823/0.0793	0.988	+0.60	0.4/0.5/0.6	Yes	None
23.00	Accepted	0.100	45.1/46.0	0.0484/0.0439	0.983	+0.60	0.2/0.4/0.6	Yes	None
23.51	Accepted	0.100	39.3/39.7	0.0894/0.0912	0.966	+0.20	0.0/0.2/0.4	Yes	None
24.00	Accepted	0.100	40.2/42.5	0.0813/0.0659	0.976	+0.80	0.4/0.6/0.8	Yes	None
24.50	Accepted	0.100	42.1/41.8	0.0655/0.0702	0.991	+0.00	0.2/0.0/0.0	Yes	None
25.00	Accepted	0.100	42.4/42.2	0.0649/0.0679	0.984	+0.40	0.2/0.3/0.4	Yes	None
25.51	Accepted	0.100	40.3/41.5	0.0784/0.0731	0.990	+0.20	0.0/0.1/0.2	Yes	None
26.00	Accepted	0.100	42.7/41.4	0.0597/0.0745	0.986	+0.40	0.2/0.3/0.4	Yes	None

Depth	Review state	Unc.	SNR L/R	Noise RMS L/R	Corr.	Lag	Delta PT/Z/M	Polarity	Warnings
26.50	Accepted	0.100	40.9/41.7	0.0736/0.0681	0.991	+0.40	0.2/0.3/0.4	Yes	None
27.00	Accepted	0.100	43.5/42.2	0.0544/0.0651	0.988	+0.20	0.0/0.2/0.4	Yes	None
27.51	Accepted	0.100	41.2/41.2	0.0703/0.0759	0.980	+0.60	0.4/0.4/0.6	Yes	None
28.00	Accepted	0.100	41.7/41.0	0.0667/0.074	0.971	+0.60	0.2/0.4/0.6	Yes	None
28.51	Accepted	0.100	43.0/41.2	0.0553/0.0752	0.988	+0.60	0.2/0.4/0.6	Yes	None
29.01	Accepted	0.100	39.2/38.7	0.0803/0.097	0.971	+0.80	0.6/0.7/0.6	Yes	None
29.51	Accepted	0.100	39.9/46.2	0.0781/0.0399	0.992	+0.40	0.2/0.2/0.2	Yes	None
30.00	Accepted	0.100	40.7/42.4	0.0649/0.0633	0.986	+1.00	0.8/1.0/1.0	Yes	None
30.52	Accepted	0.100	38.2/41.6	0.085/0.0742	0.964	+1.20	0.8/1.0/1.2	Yes	None
31.01	Accepted	0.100	42.1/41.7	0.0577/0.0692	0.996	+0.20	0.0/0.2/0.2	Yes	None
31.50	Accepted	0.100	41.6/43.1	0.0638/0.0565	0.993	+0.60	0.4/0.4/0.3	Yes	None
32.00	Accepted	0.100	40.2/41.0	0.0644/0.0707	0.985	+1.20	1.2/1.1/1.2	Yes	None
32.51	Accepted	0.100	38.2/40.7	0.0705/0.0691	0.990	+0.40	0.4/0.5/0.4	Yes	None
33.00	Accepted	0.100	37.1/41.4	0.0453/0.066	0.872	+0.60	0.4/0.4/0.6	Yes	None
33.50	Accepted	0.100	39.6/37.2	0.07/0.105	0.993	+0.40	0.2/0.4/0.4	Yes	None
34.00	Accepted	0.100	30.0/37.8	0.0842/0.0801	0.974	+1.00	0.8/0.8/1.0	Yes	None
34.51	Accepted	0.100	34.3/39.3	0.0562/0.0584	0.926	+0.60	0.1/0.3/0.3	Yes	None
35.01	Accepted	0.100	33.8/38.5	0.0891/0.0615	0.994	+0.60	0.4/0.4/0.4	Yes	None
35.51	Accepted	0.100	35.6/38.0	0.0769/0.0634	0.992	+0.20	0.0/0.0/0.0	Yes	None
36.01	Accepted	0.100	36.2/39.3	0.0668/0.0507	0.995	+0.40	0.2/0.2/0.2	Yes	None
36.51	Accepted	0.100	33.3/38.5	0.082/0.0621	0.993	+0.80	0.6/0.7/0.8	Yes	None
37.01	Accepted	0.100	34.4/36.2	0.0691/0.065	0.994	+0.00	0.2/0.1/0.0	Yes	None
37.47	Accepted	0.100	36.2/33.7	0.0607/0.062	0.978	+0.40	0.4/0.3/0.2	Yes	None

Layer velocity results

Layer	Top (m)	Bottom (m)	First peak/trough Vs	Pair crossover Vs	Individual zero crossing	Max peak (experimental) Vs	Residual (ms
1	0.00	1.01	910.5	735.7	737.0	502.8	+1.339
2	1.01	1.50	585.8	481.6	480.6	319.3	+0.549
3	1.50	2.00	404.1	338.8	337.3	221.9	-0.344
4	2.00	2.50	313.3	267.7	266.6	180.0	-1.276
5	2.50	3.01	271.5	233.9	233.6	170.3	-0.579
6	3.01	3.51	248.7	210.5	211.0	176.7	-0.154
7	3.51	4.00	225.5	185.0	186.3	185.7	+2.331
8	4.00	4.51	194.8	159.6	160.9	185.6	-1.864
9	4.51	5.00	164.1	146.5	147.1	174.8	-0.545
10	5.00	5.50	133.8	137.9	137.7	160.8	+3.129
11	5.50	6.01	116.5	134.6	134.1	156.1	-2.074
12	6.01	6.51	132.3	154.5	153.9	171.9	-0.619
13	6.51	7.00	170.1	190.0	189.3	200.2	+0.017
14	7.00	7.51	215.8	228.9	228.2	230.2	+0.274
15	7.51	8.00	253.4	258.8	258.1	253.2	+0.262
16	8.00	8.51	273.9	274.4	273.6	265.8	+0.129
17	8.51	9.00	279.4	277.6	276.7	270.2	+0.274
18	9.00	9.51	276.8	274.4	273.6	270.2	-0.108
19	9.51	10.00	272.8	270.8	270.2	269.6	-0.146
20	10.00	10.51	270.2	268.6	268.3	269.0	-0.086
21	10.51	11.00	268.4	266.5	266.9	267.0	-0.178
22	11.00	11.51	264.9	262.9	263.7	262.3	+0.190
23	11.51	12.00	258.4	256.6	257.4	255.2	+0.131
24	12.00	12.51	250.8	249.7	250.3	248.2	-0.014
25	12.51	13.01	244.8	244.4	244.7	243.4	-0.144
26	13.01	13.53	241.1	241.3	241.4	240.4	+0.042
27	13.53	14.00	239.4	239.8	239.7	238.1	+0.195
28	14.00	14.51	239.8	240.0	239.9	237.4	-0.252
29	14.51	15.00	242.7	242.3	242.4	239.1	+0.009
30	15.00	15.50	245.8	244.7	244.9	241.5	+0.023
31	15.50	16.00	247.5	245.7	246.0	243.0	+0.096
32	16.00	16.51	247.2	245.0	245.2	243.0	-0.138
33	16.51	17.01	244.7	242.7	242.7	241.5	-0.010
34	17.01	17.51	239.6	238.1	237.9	237.2	+0.098
35	17.51	18.00	232.6	231.7	231.5	230.7	+0.138
36	18.00	18.51	225.7	225.4	225.1	224.2	-0.224
37	18.51	19.00	220.1	220.5	220.3	219.3	+0.121
38	19.00	19.51	215.3	216.5	216.4	215.3	-0.091
39	19.51	20.01	212.1	213.7	213.8	212.8	+0.175

Layer	Top (m)	Bottom (m)	First peak/trough Vs	Pair crossover Vs	Individual zero crossing (experimental) Vs	Max peak (experimental) Vs	Selected residual (ms
40	20.01	20.51	210.8	212.1	212.2	211.7	-0.140
41	20.51	21.00	211.7	212.1	212.1	211.8	-0.126
42	21.00	21.51	212.9	212.6	212.5	211.9	+0.153
43	21.51	22.00	213.1	212.4	212.3	211.5	+0.032
44	22.00	22.50	213.3	212.6	212.6	211.7	-0.032
45	22.50	23.00	214.1	213.9	213.9	213.1	-0.026
46	23.00	23.51	216.2	216.4	216.5	215.6	+0.126
47	23.51	24.00	220.0	220.1	220.2	219.3	-0.092
48	24.00	24.50	225.5	225.1	225.3	224.0	-0.101
49	24.50	25.00	230.8	230.1	230.2	228.6	-0.039
50	25.00	25.51	233.4	233.0	233.1	231.2	-0.015
51	25.51	26.00	232.6	233.2	233.2	231.0	+0.199
52	26.00	26.50	230.3	231.4	231.4	229.0	+0.066
53	26.50	27.00	229.7	230.5	230.5	228.1	-0.054
54	27.00	27.51	232.5	232.3	232.3	229.8	+0.075
55	27.51	28.00	237.9	236.8	236.8	234.2	-0.214
56	28.00	28.51	244.7	243.2	243.2	240.7	-0.100
57	28.51	29.01	250.0	248.2	248.1	245.9	+0.117
58	29.01	29.51	251.2	249.2	249.1	247.3	-0.078
59	29.51	30.00	248.1	246.3	246.0	244.4	+0.118
60	30.00	30.52	241.2	239.8	239.5	238.0	+0.053
61	30.52	31.01	233.4	232.3	231.9	230.3	+0.001
62	31.01	31.50	227.1	226.0	225.6	223.4	-0.128
63	31.50	32.00	223.7	222.1	221.8	219.0	+0.211
64	32.00	32.51	223.0	221.3	221.4	218.5	-0.026
65	32.51	33.00	226.7	225.5	225.8	223.6	-0.000
66	33.00	33.50	235.8	235.1	235.5	233.8	-0.136
67	33.50	34.00	249.7	249.1	249.4	247.8	-0.018
68	34.00	34.51	265.8	264.9	265.0	262.6	-0.117
69	34.51	35.01	281.0	279.7	279.6	275.4	-0.014
70	35.01	35.51	292.6	291.0	290.6	284.9	+0.166
71	35.51	36.01	300.6	298.3	297.9	291.3	+0.191
72	36.01	36.51	307.6	304.0	303.9	297.2	-0.026
73	36.51	37.01	316.9	311.0	311.6	304.8	-0.280
74	37.01	37.47	328.7	319.8	321.2	314.2	+0.130